PALEOTRENCH — Game Plan

> Living design document. Originally the approved development plan; updated as decisions change. Current progress is tracked in the README.

1. Context

You want a first-person underwater horror game in Unreal Engine, built in this repo (`dexoisalbert-lgtm/vigilant-engine`), PC first (Steam, KB/M + gamepad), mobile later (free with ads/IAP). A scuba diver is pulled through a wormhole into a prehistoric cave system and must kill four bosses, collect shards, and repair a time machine before oxygen runs out.

Decisions locked from your answers:

Topic	Decision
Title / project name	**Paleotrench**
Scope	Full game, 8+ hours, all four bosses at first PC release
Bosses	Tusoteuthis, Livyatan, Arthropleura, Stygiomedusa — all kept, lore adapted
Boss structure	Stalk phase through the cave, then kill in a final arena
World	Central hub + 4 cave branches, any order
Caves	Fully procedural, random; seed fixed per save file
Death	Selectable difficulty: checkpoints (Normal), loot-drop (Hard), permadeath (Nightmare)
Oxygen	Air pockets (free refill + checkpoint) and spare tanks; tank upgradeable
Weapons	Retrievable + craftable ammo, special ammo from rare minerals
Power	Kill eels, harvest organ → battery pack → manually charge devices
Boss damage	Only a fully charged (level 5 of 5) eel-powered Volt Crossbow hurts bosses; on Hard the pack can't feed both head flashlight and crossbow
Mining	Slow, noisy drilling that attracts creatures
Fauna	Rich ecosystem of real species
Story	Captain's voice in fragments through static; subtitles + placeholder voice first
Tone	Graphic, Mature
HUD	Diegetic wrist dive computer (minimal overlay added for mobile)
Extras	Creature codex, multiple endings, localisation, accessibility
Art	Greybox first, real art swapped in later
You	Unreal beginner; RTX 4060/16 GB laptop (daily), RTX 5090/64 GB laptop (occasional)

Constraints that shape everything

- **I cannot run the Unreal Editor or compile Unreal code from the cloud.** I write C++, config, data files, editor-automation Python and guides; you pull, build and playtest on your laptop and send back logs/screens. So the design is **C++-first and data-driven**, with almost nothing requiring hand-made Blueprints.
- **Engine-free logic is tested by me before you ever see it.** Cave generation, oxygen/battery maths, crafting, save data and AI sense maths live in plain C++ with no Unreal headers, with a test suite I run here.
- **8+ hours, solo, beginner, realistic creatures is a very large project.** Procedural caves and systemic AI supply the hours; the plan is staged so there is a playable build early and every milestone is shippable as-is.
- **"Realistic looking" is gated by art, not code.** Greybox gets the game fun and scary first; section 11 covers the art swap.
- **Mature gore + free-with-ads on mobile conflict**: ad networks and stores restrict graphic content. The mobile build ships with reduced gore by default and age gating (section 12).

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2. Step 0 — Repo reset (needs your go-ahead when we start)

The repo currently contains a complete Roblox Luau plane game (7 commits, 41 files: `src/`, `docs/GAME_PLAN.md`, `rokit.toml`, etc.).

1. Done: the Roblox project was removed; it stays recoverable from git history (last commit `32886ff`).
2. `git rm` everything; commit "Remove previous project".
3. Add Unreal `.gitignore`, `.gitattributes` with **Git LFS** for `*.uasset`, `*.umap`, audio, textures.
4. All work goes on branch `ccr-c50880a2-g0siyj`; no PR unless you ask.

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3. Your machine setup (guide I'll write: `Docs/SETUP.md`)

- Epic Games Launcher → **latest stable Unreal Engine 5** (version pinned in `Paleotrench.uproject`; both laptops must match).
- Visual Studio 2022 with "Game development with C++" workload; Git + Git LFS.
- Later, for mobile: Android Studio/SDK/NDK via Unreal's `SetupAndroid` script. iOS builds need a Mac — flagged as a future decision.
- Daily loop: `git pull` → double-click `Tools/Build.bat` → open project → play → run `Tools/CollectLogs.bat` and send me the output if anything breaks.
- 4060 laptop = daily development and the **minimum-spec benchmark**. 5090 laptop = lighting builds, packaging, high-settings capture.

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4. Technical architecture

Repo layout

```
Paleotrench.uproject
Config/                  DefaultEngine/Game/Input/Scalability + Android/iOS device profiles
Source/
  PaleotrenchCore/       engine-free C++ (std only): cave graph, resource maths, crafting,
save schema
  Paleotrench/           game module: player, gear, weapons, AI, bosses, world, UI, audio,
save
  PaleotrenchEditor/     editor utilities: greybox asset bootstrap, data validation
Content/                 binary assets (LFS) — created by scripts or by you in-editor
Data/                    CSV/JSON source of truth for all tuning (imported to DataTables)
Tools/                   Build.bat, CollectLogs.bat, CoreTests (CMake), Python editor scripts
Docs/                    SETUP, GDD, per-milestone playtest checklists, art briefs
```

Key technical choices

- **C++ for all gameplay; Blueprints only as thin data-only subclasses** generated by a Python editor script, so you never wire graphs by hand.
- **Everything tunable lives in `Data/*.csv`** (creature stats, weapon damage, oxygen rates, upgrade costs, spawn tables) so balance changes need no code.
- **Swimming**: custom 6-degrees-of-freedom movement mode on `UCharacterMovementComponent` — inertia, drag, buoyancy, slow turn while sprinting, fin-kick noise events. No simulated water surface except air pockets and the boat intro.
- **Underwater look**: exponential height fog + distance-based colour absorption post-process, Niagara marine snow and silt, caustics light function near openings, mask-edge vignette and droplets. Fully dynamic lighting (procedural levels cannot be baked): flashlight spotlight + emissive bioluminescence. Lumen is a High/Ultra option only; the default path is cheap enough for mobile.
- **AI**: State Trees per creature, `UAIPerceptionComponent` with custom senses (below), Environment Query System for hiding/ambush spots, 3D navigation via a sparse voxel graph generated with the cave (NavMesh does not work for swimming volume).
- **Unified stimulus system** — the heart of the horror. Every player action emits typed stimuli that creatures sense differently:
  - `Sound` (fin kicks, breathing bubbles, drill, weapon fire, radio)
  - `Light` (flashlight level, flares, bioluminescence)
  - `Vibration` (touching rock, mining, walking in air pockets)
  - `Electric` (charging devices, eel discharge)
  - `Blood` (wounds, corpses — spreads over time)
- **Input**: Enhanced Input with an abstract action layer (KB/M, gamepad now; touch later with no gameplay code changes).
- **Save**: versioned `USaveGame` holding seed, hub state, inventory, upgrades, boss flags, codex, checkpoints. Steam Cloud on PC.
- **UI**: UMG widgets driven from C++ (CommonUI for gamepad/touch focus). String Tables for every line of text from day one (localisation).
- **Audio**: MetaSounds; underwater low-pass, occlusion via cave graph, sound-class ducking for breath/heartbeat.

- **Scalability**: Low/Medium/High/Ultra tiers defined at the start; "Low" is the future mobile profile.

Procedural cave generation (seed fixed per save)

1. **Graph** (engine-free, tested here): per-cave seed = save seed + cave ID → nodes (chambers, air pockets, mineral rooms, nests, set-piece rooms, boss arena) and edges (tunnels, squeezes, vertical shafts, loops for escape routes). Rules guarantee: solvable, air pocket spacing within oxygen range for the current tier, at least two routes around choke points, arena furthest from entrance.
2. **Geometry**: tunnels = noise-displaced tubes swept along splines; chambers = signed-distance blobs; meshed in chunks (dynamic mesh), collision generated async, streamed by distance.
3. **Set-piece templates**: hand-authored room templates (boss arenas, the hub, story rooms, special scares) slot into graph nodes via sockets. Authored in C++/data for greybox; replaced by art-dressed level instances later.
4. **Population**: spawn tables per biome for fauna, eels, mineral nodes, shards, logs, ammo, hiding spots (kelp-like growths, crevices, shipwreck debris from other eras).
5. **Scare director**: places scripted-feeling moments procedurally (silhouette passing a tunnel mouth, radio burst, light flicker, corpse drift) using pacing rules — tension builds, releases, never repeats twice in a row.
6. **Debug**: seed displayed in pause menu; `Tools/cavegen` CLI prints an ASCII/SVG map of any seed so I can reproduce your bug reports here.

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5. Story and structure

- **Prologue — The boat.** Dawn, small research vessel. Radio check with Captain (tutorial for radio, wrist computer, flashlight charging). Routine trench dive: swimming, light levels, a calm modern reef for contrast.
- **The Rift.** At depth, a distortion in the trench wall pulls you through. Gear glitches, the captain's voice stretches into static.
- **The Confluence (hub).** A vast cavern where time strata collide — which is why creatures of different eras coexist (Carboniferous Arthropleura, Cretaceous Tusoteuthis, Miocene Livyatan, and Stygiomedusa as an ancient lineage that never died out). Here: a wrecked time machine left by an earlier expedition, a workbench, an air dome, storage, and four cave mouths.
- **Goal.** The machine needs **4 Core Shards** (one inside each boss's lair, released on its death) plus **minor shards** scattered through caves (each cave has more than required; extras feed endings and upgrades).
- **Radio.** The captain reaches you in fragments. Using the radio costs battery and makes sound. Time runs differently: his messages arrive out of order, and logs from earlier divers suggest he has sent people down before.
- **Endings** (driven by shards, logs found, and a final choice):
    1. *Return* — minimum repair; you surface, something came with you.
    2. *True Return* — all minor shards; clean jump, captain confronted.
    3. *Stay/Seal* — destroy the machine to close the rift.
    4. *Lost* — die during the final escape sequence on Nightmare.

- **Finale.** Activating the machine triggers a collapse and a timed escape swim with every surviving creature agitated.

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6. Core gameplay systems

Oxygen (the clock)

- Tank drains continuously; faster when sprinting, panicking (near bosses), or wounded. Low oxygen: tunnel vision, loud heartbeat, slowed movement, then blackout.
- Refill free at **air pockets** (also checkpoints, safe-ish rooms where you can stand) and at the hub. **Spare tanks**: found or crafted, swapped manually with an animation that leaves you vulnerable.
- Upgrades: capacity, rebreather efficiency, emergency pony bottle.

Electricity

- **Battery pack** holds charge; **flashlight, radio, wrist computer scanner, electric ammo** each have their own small cell.
- Charging is manual: hold to plug the device into the pack (hands occupied, emits `Electric` stimulus, takes seconds).
- **Flashlight** brightness levels: Off / Low / Medium / High / Burst (strobe stun, heavy drain). Higher levels reveal more and attract more. Flicker when cell is low. With the light off, the scare director gets bolder and eyes/shapes appear at the edge of vision.
- **Eels** (a fictional prehistoric electric species, e.g. "voltaic eel", presented as an undiscovered lineage): hunt in crevices, shock on contact (damage + drains devices + scrambles wrist computer). Kill → harvest **electric organ** (short gory interaction) → refills pack. Organs decay if carried too long. A weak hand-crank fallback on the pack prevents soft-locks (slow, noisy).

The Volt Crossbow — the only boss-killer (your addition)

- The crossbow is an **electric weapon charged from eel power**, with **5 charge levels** shown as five lit segments on the weapon itself and on the wrist computer.
- Charging is manual from the battery pack, one level at a time: hold to charge, hands occupied, slowed, emits `Electric` + `Sound` stimuli. Each level takes longer and costs more pack charge than the last. Being hit interrupts the level in progress.
- Damage by level: 1–2 kill small fauna, 3–4 kill/stagger large fauna and mini-bosses, **only level 5 (full charge) damages a boss**. Anything less only makes a boss flinch. A shot always empties the crossbow back to 0.
- **Every boss kill is therefore a loop**: expose the weak point → fire a full-charge bolt → survive while recharging to level 5 with the boss enraged → repeat. Each boss needs several full-charge hits (3 on Normal, more on Hard/Nightmare). All other weapons and tools exist to stun, distract, expose weak points and buy charging time.
- Eel organs are the fuel, so boss arenas contain eel nests: you must kill eels mid-fight to keep the pack fed.
- **Charge vs. light dilemma**: the pack has one output — it charges the head flashlight *or* the crossbow, never both at once.

    - Normal: pack capacity is enough to keep both topped up if you harvest sensibly; the cost is time.
    - **Hard**: one eel organ cannot fully charge both. You must choose — see, or kill. Fights have stretches in darkness by design.
    - Nightmare: as Hard, plus charge leaks slowly from both devices.
- Upgrades: faster charging, charge retention, a capacitor that holds one spare level, quieter charging. No upgrade removes the level-5 rule.
- The flashlight is **head-mounted** (mask light), so it stays on while both hands are busy charging or aiming.

Other weapons (support — cannot damage bosses)

Weapon	Role	Notes
Dive knife	Start; harvesting, last resort	Used for eel organs and escaping grabs
Hand spear	Reusable melee/thrust	Upgradable tip; can be thrown and retrieved
Speargun (harpoon)	Heavy single shot	Slow reload; tether upgrade lets you reel the shaft back or pin creatures
Bang stick / powerhead	Contact explosive	Rare, very loud
Flares	Light + distraction	Lure light-seeking creatures away
Noise lures	Thrown sound decoy	Key against echolocators

- Projectiles have drag and drop; shafts stick in creatures/rock and must be **swum to and pulled out**.
- **Crafted ammo** at hub workbench from minerals: standard, barbed (bleed), explosive tip, electric (uses charge), tranquiliser toxin (from fauna glands).
- Hit locations and weak points per species; armour deflects.

Mining and upgrades

- **Drill** nodes over several seconds: emits `Sound` + `Vibration`, kicks up silt (blinds you), costs oxygen time. Can be interrupted and resumed.
- Minerals by rarity and depth (e.g. pyrite, quartz, magnetite, amber, "chronolite" rare). Carry limit; on Hard you drop the dive's haul on death.
- **Upgrade trees** at workbench: Suit (armour, thermal, silent fins), Tank, Light (efficiency, beam, UV mode that reveals trails), Pack (capacity), Radio (range, sonar ping), Weapons, Drill (quieter, faster), Wrist computer (map, scanner, threat indicator).

Wrist dive computer (diegetic HUD)

Raise wrist to see oxygen, depth, pack and cell charge, compass, auto-map of explored graph, objectives. Peripheral cues without raising it: breath sound, a single mask-edge LED for oxygen and battery. Scanner mode fills the **Codex**.

Health and injuries

Wounds bleed (emits `Blood` stimulus that draws predators), limb injuries slow swimming or shake aim, treated with limited med kits at air pockets. Graphic first-person death sequences per

creature, with a reduced-gore setting.

Difficulty

Normal (checkpoint respawn), Hard (hub respawn, loot dropped at body, scarcer supplies), Nightmare (permadeath, no map threat indicator). Accessibility assists are separate from difficulty.

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7. The four bosses

Each cave has: a biome, a stalk phase where the boss roams the generated cave and reacts systemically, **three "exposure" objectives** that weaken it or reveal its weak point, then an authored arena fight in three phases. Killing it releases a Core Shard.

Common kill rule: in every phase below, the described action (stun, lure, sever, burn, pin) only *opens the weak point*; the actual damage is one **level-5 Volt Crossbow bolt** into it. Then the boss enrages while you recharge from 0 to 5, harvesting arena eels for power. The per-boss sense rules make recharging uniquely dangerous in each fight: charging is noisy for Livyatan, visible sparks for Tusoteuthis, an electric beacon for Stygiomedusa, and keeps you standing still on rock for Arthropleura.

Livyatan melvillei — "The Drowned Cathedral" (huge open flooded caverns, whale-fall graveyards)

- **Sense**: blind; echolocation. Periodic click trains sweep the cave — anything moving fast during a sweep is "seen". Hears sound acutely.
- **Player rule**: move slowly and silently, freeze during clicks, use noise lures. Flashlight is safe; drilling is deadly.
- **Stalk**: patrols main galleries (too big for tunnels — tunnels are your refuge, but it rams walls causing cave-ins).
- **Exposure**: recover three resonator parts from an old expedition to build a sonic disruptor.
- **Arena**: phase 1 jam its sonar and strike the melon organ when stunned; phase 2 it thrashes blind, bring down stalactites; phase 3 inside-the-jaws grab sequence, harpoon the wounded organ.

Tusoteuthis longa — "The Ink Galleries" (tight vertical shafts, kelp-like forests, drifting ink clouds)

- **Sense**: deaf; exceptional eyesight, tracks motion and light. Camouflages against rock.
- **Player rule**: break line of sight, hide in growths and crevices, lights off or low, move when it looks away. Noise is free here — mine all you want.
- **Stalk**: ambusher; hooked arms reach into tunnels and feel around; ink clouds blind you.
- **Exposure**: destroy three camouflage "nest" sites and craft UV light mode to see through its camouflage.
- **Arena**: phase 1 sever feeding tentacles when they grab; phase 2 fight in ink with UV; phase 3 it drags you toward its beak — spear the eye/beak.

Stygiomedusa gigantea — "The Black Veil" (pitch-dark abyssal chambers, bioluminescent fauna)

- **Sense**: no eyes or ears; drawn to light and electricity. Four 10 m oral arms billow through whole rooms like curtains.
- **Player rule**: flashlight discipline in the darkest cave of the game; charging devices is dangerous; use flares to pull the veil elsewhere.
- **Stalk**: slow, silent, everywhere — arms drift through tunnels; contact envelops and digests (oxygen drains fast, struggle to escape).
- **Exposure**: collect three thermal-vent catalysts to craft incendiary/electric overload ammo; learn it retreats from vent heat.
- **Arena**: phase 1 lure arms into vents to burn them; phase 2 lights-out — navigate by bioluminescence to reach the bell; phase 3 overload it with a full battery-pack discharge (leaving you powerless for the escape).

Arthropleura — "The Breathing Tunnels" (half-flooded maze with long air-filled crawlways)

- **Lore**: an air-breather that hunts the air pockets — the one place you thought was safe. From this cave on, checkpoints feel different.
- **Sense**: poor sight and hearing; feels **vibration through rock** and tastes blood.
- **Player rule**: in air sections you walk/crawl — every footstep, dropped item and drill hit carries. Crouch-walk, avoid loose gravel, slip back into water (it cannot swim well, but it can hold its breath and follow briefly).
- **Stalk**: 2.5 m armoured body pours through crawlways and across ceilings; you hear its legs before you see it.
- **Exposure**: find three soft-joint weak points via codex scans of moulted shells; craft armour-piercing tips; flood controls for the arena.
- **Arena**: phase 1 hit underbelly when it rears; phase 2 flood/drain the chamber to control where it can go; phase 3 pin it with tethered harpoons and finish at the head.

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8. Ecosystem (real species, data-driven)

All defined in `Data/Creatures.csv` using shared behaviour archetypes, so adding a species is mostly a data row plus a mesh.

- **Hostile**: Jaekelopterus and other eurypterids (sea scorpions, ambush from silt), Helicoprion (whorl-tooth, fast pack hunter), Xiphactinus (charging fish), Dunkleosteus (armoured mini-boss guarding rare minerals), Rhizodus, Mosasaur juvenile (open-water mini-boss), voltaic eels.
- **Territorial/neutral**: Orthocone nautiloids (Cameroceras), Anomalocaris, Archelon (giant turtle), Stethacanthus.
- **Harmless / atmosphere / false scares**: ammonites, trilobites, belemnite shoals, crinoid "meadows", Hallucigenia, Tullimonstrum, bioluminescent plankton.
- **Systemic behaviour**: predators hunt prey and each other, corpses attract scavengers, blood spreads, shoals scatter when a boss approaches (an early-warning system the player learns to read).

- **Codex**: scan to unlock entry with real facts (era, size, diet, fossil sites) + gameplay hints. Clearly labels the two fictional liberties (voltaic eel; where each animal really lived).

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9. Horror direction

- **Sound first**: breathing regulator, heartbeat, muffled distance, creature vocalisations placed by cave graph. Radio static intensifies near bosses and bursts with half-words.
- **Darkness as mechanic**: light reveals and attracts; turning it off invites the scare director.
- **Scare director**: tracks a tension value; schedules ambient, near-miss and direct scares with cooldowns so procedural caves still feel authored.
- **Never fully safe**: air pockets are safe until the Arthropleura cave teaches otherwise; the hub is breached once late-game.
- **Graphic content**: per-creature death sequences, eel organ harvest, whale-fall carcasses, previous divers' remains. Reduced-gore toggle swaps blood for dark silt and cuts away from deaths.
- **Accessibility**: subtitle size/background, colour-blind palettes, camera shake/bob sliders, hold-vs-toggle, aim assist (gamepad), arthropod-phobia mode (reshapes Arthropleura and eurypterids), reduced-intensity mode for thalassophobia, audio cue visualiser.

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10. Development roadmap

Each milestone ends with: my engine-free tests passing, a build you compile, a short playtest checklist in `Docs/Playtests/`, and a pushed commit.

#	Milestone	What you can do at the end
M0	Repo reset, project skeleton, setup guide, build/log scripts, CoreTests harness	Project opens and compiles on your laptop
M1	Swimming, camera, input (KB/M + gamepad), underwater look, test tank level	Swim around a greybox tank; feels like diving
M2	Oxygen, battery pack, flashlight levels, manual charging, wrist computer, save/load	Survive on a timer, manage light
M3	Cave generator: graph → geometry → streaming, 3D nav, debug map CLI	Explore an endless variety of seeded caves
M4	Stimulus system, creature AI archetypes, eels + 4 lesser species, health/injury	Caves are alive and hunting you
M5	Weapons, Volt Crossbow 5-level charging, retrievable ammo, mining, minerals, workbench, crafting, upgrades	Full core loop
M6	Hub, time machine, shards, checkpoints, difficulty modes, cave branch travel	**Vertical slice** structure: hub → cave → back
M7	Boss 1 Livyatan (stalk AI, exposure objectives, arena)	First complete cave
M8	Boss 2 Tusoteuthis	
M9	Boss 3 Stygiomedusa	

#	Milestone	What you can do at the end
M10	Boss 4 Arthropleura + air-section movement	All four caves
M11	Prologue boat, rift sequence, radio dialogue system, logs, endings, finale escape	Start-to-credits playable
M12	Scare director, full ecosystem roster, codex, audio pass	Content complete (greybox)
M13	Menus, settings, accessibility, localisation pipeline, gamepad polish	Feature complete
M14	Art swap (section 11), lighting, VFX, performance pass on 4060 laptop	Beta
M15	Steam: Steamworks, achievements, cloud saves, store page assets, demo, playtests, bug fixing	**PC release**
M16+	Mobile port (section 12)	

Order within M7–M10 can change; each boss is isolated behind a common `ABossBase` + per-boss State Tree and data.

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11. Art and audio pipeline (greybox → final)

- Greybox uses primitive/capsule creatures with correct size, hitboxes and sockets, generated by `PaleotrenchEditor` so no manual asset work is needed to play.
- Every creature references assets through a **data asset slot** (mesh, anim blueprint, sounds, VFX). Swapping art = changing a row, no code.
- `Docs/ArtBriefs/` — one brief per creature: scale, paleontological references, required animations (swim, turn, attack set, hit, death, idle), bone/socket names, poly and texture budgets for PC and mobile LODs.
- Sources you can choose per asset later: Fab/Megascans (rock, coral, debris), commissioned or purchased rigged creatures, AI-assisted generation cleaned in Blender. Bosses deserve custom or high-quality purchased rigs; they sell the game.
- Procedural swim animation (spine-chain bending driven by velocity, Control Rig) reduces hand-animation needs for fish-like and tentacled creatures, and looks more natural in a random cave.
- Audio: placeholder library sounds → final pass; voice: subtitles + synthetic placeholder → actors for Captain and logs.

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12. Mobile port plan (post-PC)

Prepared from day one: abstract input, scalability tiers, dynamic-only lighting, chunk streaming, data-driven asset slots with mobile LODs.

- **Rendering**: Mobile forward renderer, Vulkan (Android) / Metal (iOS), no Lumen/Nanite, cheaper fog and post-process, fewer dynamic lights (flashlight + 2–3), smaller streaming radius, lower creature counts via spawn-table multipliers, 30 fps target on mid-range devices.

- **Controls**: twin virtual sticks + ascend/descend buttons, gyro aim option, contextual interact button, aim assist, larger wrist-computer UI, optional minimal HUD overlay.
- **Sessions**: more frequent autosave at air pockets; suspend/resume safe.
- **Monetisation (free + ads/IAP)**, designed not to break horror: no banner ads in gameplay; optional rewarded ad for a checkpoint revive or a supply cache at the hub; one-time IAP removes ads; cosmetic gear skins; possibly first cave free with full-game unlock. No selling oxygen or power.
- **Store compliance**: reduced gore default, age gate, 17+/Mature rating, ad network content categories checked early; iOS requires a Mac for builds.
- **Size**: target under store cellular limits with on-demand asset packs per cave.

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13. Risks

Risk	Mitigation
I can't compile here; errors only show on your machine	Engine-free core tested here; small commits; `Build.bat` + `CollectLogs.bat` round-trip; strict use of stable, well-documented engine APIs
Scope (8+ h, 4 bosses, ecosystem)	Systems + procedural content provide hours; milestones M6 and M7 are natural early-access/demo points if needed
Random caves weaken authored horror	Scare director, authored set-piece templates, authored boss arenas
Procedural mesh + dynamic light performance	Chunked async generation, aggressive streaming, budget tested on the 4060 laptop every milestone
Realistic art for unusual animals is hard to source	Art briefs, procedural animation, bosses prioritised for budget
16 GB RAM laptop with UE5 C++ builds	Unity builds tuned, small modules, heavy tasks on the 5090 laptop
Mature + ads on mobile	Reduced-gore mobile default, premium unlock path as fallback

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14. Verification

- **Here (every commit)**: `Tools/CoreTests` (CMake + g++) — cave graphs for thousands of seeds are connected, solvable and within oxygen range; resource, crafting and save round-trip tests; `cavegen` map output for visual inspection.
- **On your laptop (every milestone)**: `Tools/Build.bat` compiles; Unreal automation tests run from a script; you follow the milestone checklist in `Docs/Playtests/Mx.md` (specific things to try, expected results, what to screenshot) and send logs via `CollectLogs.bat`.
- **Performance**: `stat unit` / `stat gpu` captures on the 4060 laptop against per-tier frame budgets; seeds of any bad cave recorded for reproduction.
- **Before release**: full playthroughs on all three difficulties, all endings reached, gamepad-only run, accessibility options checked, localisation pseudo-language pass.

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15. First actions once you approve

1. Step 0 repo reset (tag archive, wipe, LFS, ignore files).
2. M0: `Paleotrench.uproject`, three modules, configs, `Docs/SETUP.md`, build/log scripts, CoreTests harness, `Docs/GDD.md` (this plan, expanded as the living design document).
3. You install Unreal + Visual Studio following `SETUP.md` and confirm the empty project compiles — then M1 begins.